Literature status: frequent disjoint hypercyclicity need not be dense

Automated mathematics run fa_banach_001, lane 2

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Status: literature already answered; both clauses of Question 2.6 have negative answers.

Source question

Oezguer Martin and Rebecca Sanders, *Extending Families of Disjoint Hypercyclic Operators*, arXiv:2312.07054, Question 2.6 on physical PDF page 6, ask:

> Are frequently d-hypercyclic operators also densely d-hypercyclic? Do frequently d-hypercyclic operators satisfy the Disjoint Blow-up/Collapse property?

Supporting answer

Frederic Bayart, Sophie Grivaux, Etienne Matheron, and Quentin Menet, *Hereditarily Frequently Hypercyclic Operators and Disjoint Frequent Hypercyclicity*, arXiv:2409.07103. Section 7 restates the first clause as Question 7.1 and says that the next theorem solves it. Theorem 7.2, on physical page 28, states that there are a Banach space X and $T_1, T_2 \in \mathfrak{L}(X)$ such that (T_1, T_2) is d-frequently hypercyclic but not densely d-hypercyclic. Its proof is completed on physical page 32. Hence the first clause is answered negatively.

The same example answers the second clause. As recorded in the source paper's implication diagram immediately before Question 2.6, the Disjoint Blow-up/Collapse property implies dense d-hypercyclicity. Therefore the pair in Theorem 7.2 cannot have the Disjoint Blow-up/Collapse property.

Provenance and scope

This is an exact literature resolution, not a new counterexample from the run. The supporting paper explicitly formulates the same density question, labels Theorem 7.2 as its solution, and cites the source preprint arXiv:2312.07054 as reference [39]. The negative answer to the Blow-up/Collapse clause is the immediate contrapositive of the implication already stated by the source.

Thus both clauses of Question 2.6 are settled negatively. The source's Questions 1.2 and 1.3 are outside this packet because the source itself answers them later; those signals are same-paper extraction false positives.

Evidence

The packet contains the original arXiv PDF as `source_paper.pdf` and the decisive later PDF as `supporting_paper_2409.07103.pdf`. The source question is on physical page 6. The supporting question and theorem are on physical page 28, and the proof concludes on physical page 32.

References

[1] O. Martin and R. Sanders, *Extending Families of Disjoint Hypercyclic Operators*, arXiv:2312.07054 (2023).

[2] F. Bayart, S. Grivaux, E. Matheron, and Q. Menet, *Hereditarily Frequently Hypercyclic Operators and Disjoint Frequent Hypercyclicity*, arXiv:2409.07103 (2024).